

This formula for the entropy cannot be extended analytically beyond $e = (1 + \epsilon)/2$, as $\partial^2 s / \partial e^2$ diverges as $e \rightarrow (1 + \epsilon)/2$. However, $e = (1 + \epsilon)/2$ corresponds to $t = (\epsilon^3 + 3\epsilon)/4$, or, using the more basic variable e ,

$$(42) \quad t = [(2e - 1)^3 + 3(2e - 1)]/4,$$

which is in the interior of (e, t) space. (Since the graphon $g(x, y)$ is nowhere zero, it differs in form from the graphons describing graphs with minimal t .) Thus $s_{e,t}$ must fail to be analytic in some neighborhood of the first scallop. $\square$

Of course there must also be a phase transition, presumably from this bipartite phase to a homogeneous phase, if one fixes e and raises t , which we see as follows.

Corollary 5.3. *Assuming our perturbative solution is the global optimizer, there is a phase transition as one raises t , for any fixed $0 < e < 1/2$.*

Proof. Recall from Corollary 5.2 the connection between t and ϵ :

$$(43) \quad t = \epsilon^3 - (e - \epsilon)^3.$$

Note that t is an increasing function of ϵ and reaches the value $2e^3$ when $\epsilon = 2e$. From equation (41) for the entropy we see that it cannot be extended analytically to $t > 2e^3$, yet for $e < 1/2$ we have $2e^3 < e^2 < e^{3/2}$ so (e, t) is in the interior of the phase space (Figure 1). $\square$

6. CONCLUSION

Our goal was to analyze possible phase transitions between multipartite phases of complex networks, analogous to solid/solid transitions in materials. To this end we adapted the Strauss model [St], defined in the grand canonical ensemble, to a microcanonical ensemble. It is appropriate at this point to review the ‘equivalence of ensembles’ in statistical physics.

In thermodynamics the concavity of the entropy $S(E, N, V)$, as a function of internal energy E , particle number (or mass) N and volume V , and the interpretation of equilibrium states as states maximizing the entropy, are both fundamental; see for instance [Ca, Ma]. Lagrange multipliers can be used to convert this optimization criterion of the entropy to an equivalent optimization criterion of the free energy, where the free energy is the Legendre transform of the entropy [Do, Ca, Ma]. It is important that the Legendre transform between the entropy and free energy be invertible so the two optimization schemes are equivalent, and this follows from the concavity of the entropy. (See Section 26 of [Ro] for the mathematics of the Legendre transform between convex functions.)

Statistical mechanics supplies a model for thermodynamic states, as probability distributions on mechanical multiparticle states. From a given short range particle interaction one can then (in principle) compute the internal energy E and entropy S , and prove the above two features of the entropy: its concavity and its optimization role for equilibrium states. To do this one uses the basic Boltzmann/Gibbs ansatz: that the entropy $S(E, N, V)$ is proportional to

$$(44) \quad - \sum_j \rho_j \ln(\rho_j)$$

where ρ_j is the probability of multiparticle state j , and the equilibrium state is that probability distribution $\{\rho_j\}$ on the set $\Xi(E, N, V)$, of multiparticle states of energy E and particle number N in volume V , which maximizes the entropy [R1, R2, Ge, Wi, Ma]. (Note that in taking an infinite volume limit, which we must do to obtain equivalence of ensembles, one can divide the entropy's variables by volume, and consider the entropy density as a function of particle and energy density).

The equivalence of ensembles in statistical mechanics is basically a strengthening of the equivalence in thermodynamics between entropy $s_{e,t}$ and free energy ψ_{β_1, β_2} , corresponding to Lagrange multipliers β_1 and β_2 , which follows from the concavity of the entropy. With the modeling of the thermodynamic states this now implies a bijection $(e, t) \longleftrightarrow (\beta_1, \beta_2)$ such that $s_{e,t}$ and ψ_{β_1, β_2} have the same optimizing states, at least off some manageable sets of parameter pairs corresponding to ‘phase coexistence’ where the bijection can degenerate to a many-to-one map [Ge].

In exponential random graph models, which are mean field rather than short range, the entropy need not be concave [TET] and indeed this fails in an obvious way for the specific model we are analyzing, the Strauss model, since even the domain R of the entropy is not convex (see Figure 1). Therefore in the infinite node limit of the model the free energy density ψ_{β_1, β_2} , need not be equivalent to the entropy density $s_{e,t}$; ψ_{β_1, β_2} can be obtained from $s_{e,t}$ by Legendre transform, but it may not be possible to recover $s_{e,t}$ from ψ_{β_1, β_2} . Inequivalence can result from the existence of graphons maximizing $s_{e,t}$ for some (e, t) which are not maximizers of ψ_{β_1, β_2} for any (β_1, β_2) . Specific instances of such loss of information in ψ_{β_1, β_2} are shown in a future paper [RS], but one consequence can already be seen in the transitions, studied previously [PN, CD, RY] in the grand canonical ensemble, between independent-edge graphs across a phase transition curve in the phase space; see Figure 2 for the Strauss model. Such ‘free particle’ graphs, with only edge density e as a variable, optimize ψ_{β_1, β_2} for (β_1, β_2) throughout the upper half of the grand canonical phase space, so e is a function of (β_1, β_2) off the transition curve there; see [RY] for details. These graphs all lie on the curve $t = e^3$ in Figure 1, not a 2-dimensional region in that microcanonical phase space, making it difficult to use singularities of the free energy ψ_{β_1, β_2} to imply singularities of the entropy $s_{e,t}$. For this reason we have focused here on phase transitions in the lower region of the microcanonical phase space, Figure 1.

We have shown that our (2-parameter, bipartite) graphons g of Theorem 5.1 maximize the entropy density at least to second order in perturbation theory, among graphons with a limited range of edge and triangle densities. Assuming the g are actually global maximizers we then proved the entropy density would have to lose analyticity as the edge density of the graphon approaches the tripartite regime. We also show that the entropy density must suffer a phase transition as the triangle density is raised sufficiently high, presumably from the structured bipartite phase to a homogeneous phase of higher triangle density.

We expect that a more complicated analysis could produce appropriate graphons $g^{(k)}$, $k \geq 1$, near each of the higher edge density (multipartite) graphons of minimal triangle density, with a transition near each scallop intersection. Intuitively this suggests a mechanism whereby as edge density is increased, near the minimum triangle density graphon, the system progressively transitions through finer and finer structure; for high edge density most graphs would consist of many interacting ‘parts’.

Our results on phase transitions require that the graphons of Theorem 5.1 be in fact global, not just local, maximizers of the entropy density. In a future paper [RS] we use a

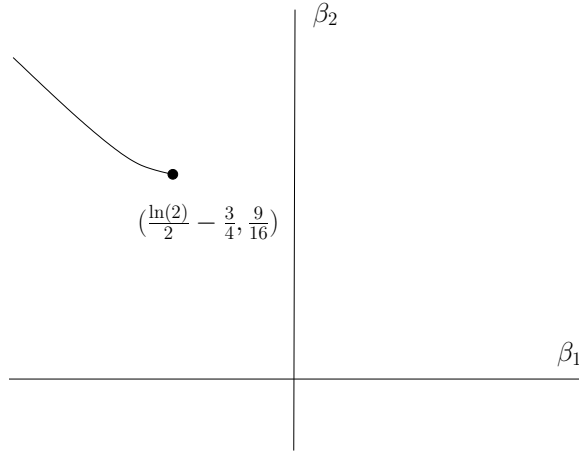


FIGURE 2. The curve of all singularities of ψ_{β_1, β_2} for $\beta_2 > -1/2$

symmetry to prove that these graphons are indeed the unique global maximizers at least for triangle density in the range $0 \leq t \leq 1/8$ and edge density $e = 1/2$, and we can then see a transition on this curve. However we still cannot prove the graphons are the global optimizers of entropy density for (e, t) in any two-dimensional region, as is needed to fully justify the notion of a structured phase. (See [AR] for a variant of this approach.)

In conclusion we emphasize that our key tool was Theorem 3.1, an optimization formula for the asymptotic entropy density, and made essential use of the graph limit formalism. The graphon formalism is a powerful tool for dealing with the infinite size limit in mean field models, and we have used it to make some progress on understanding the structure of asymptotically large graphs near the extreme of low triangle density.

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